APPM 4600 HW8 P.1

1. Hermite Hoperations

Where does Hermite come from?

The Hermite polynomial interpolating $\{(x_j, f(x_j)), (x_j, f'(x_j))\}_{j=0}^{n}$ can be expressed by

$$p_{2n+1}(x) = \sum_{j=0}^{n} f(x_j)\, Q_j(x) + \sum_{j=0}^{n} f'(x_j) R_j(x)$$

where $Q_j(x) = [1 - 2(x-x_j)\ell_j'(x_j)]\ell_j^2(x)$, $R_j(x) = (x-x_j)\ell_j^2(x)$

a. The conditions are that:

when $i \neq j$, $Q_j(x_i) = 0$ $\quad Q_j'(x) = 0$ $\quad R_j(x) = 0$ $\quad R_j'(x) = 0$

when $i = j$ $\quad Q_j(x_i) = 1$ $\quad Q_j'(x) = 0$ $\quad R_j(x_i) = 0$ $\quad R_j'(x_i) = 1$

or more simply,

$$Q_j(x_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$Q_j'(x_i) = 0 \ \forall x_i$$

$$R_j(x_i) = 0 \ \forall x_i$$

$$R_j'(x_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

b. Do they satisfy these conditions?

$Q_j(x_i) = [1 - 2(x_i - x_j)\ell_j'(x_j)]\ell_j^2(x) \rightarrow$ $i = j$, $Q_j(x_i) = 1 - 0 = 1$ ✓

$i \neq j$, $Q_j(x_i) = [1 - 2(x_i - x_j)\ell_j'(x_j)] \cdot \ell_j^2(x_i) = 0$ ✓

$Q_j'(x_i) = [-2 \cdot \ell_i'(x_j) \cdot \ell_j^2(x)] + [1 - 2(x_i - x_j)\ell_i'(x_j)] \cdot 2\ell_j'(x_i)\ell_j(x_i)$ $\quad i \neq j$, $\ell_j(x_i) = 0$ so $Q_j'(x_i) = 0$ ✓

$i = j$, $\ell_j(x_i) = 1$ so $-2\ell_i'(x_j) + 2 \cdot \ell_i'(x_j) = 0$! ✓

$R_j(x_i) = (x_i - x_j) \cdot \ell_j^2(x_i) = 0$ $\quad i = j$ b/c $x_j - x_j$

0 $\quad i \neq j$ b/c $\ell_j^2(x_i) = 0$

$R_j'(x_i) = (1)\ell_j^2(x_i) + (x_i - x_j) \cdot 2\ell_j'(x_i)\ell_j(x_i)$ $\quad i = j$, $= 1 \cdot (\ell_j^2(x_i)) = 1$ ✓

$i \neq j$, $= 0 + 2(0) = 0$ ✓

all conditions verified!

APPM 4600: HW8 P.2

2. Interpolation Adventures

Interpolate $F(x) = \frac{1}{1+x^2}$ on $[-5,5]$. Using equispaced nodes w/ $n = 5, 10, 15,$ and 20, interpolate using:

a. Lagrange
b. Hermite
c. Natural Cubic
d. Clamped Cubic.

Which performs best? why?

~~a. Lagrange:~~

a. For $n=20$ Lagrange and Hermite on equispaced nodes suffer from increasing error near either side of our interval. Splines perform better in terms of absolute error by as much as 6 orders of magnitude in the extremes. Near the center they provide comparable performance.

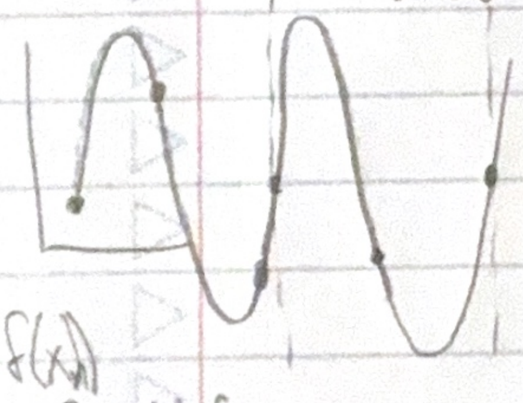
b. Chebyshev nodes drastically increase performance of Lagrange / Hermite near the sides, but appear to have little impact on spline probably b/c splines are defined locally so don't see end effects.

For $n=5$, hermite/Lagrange and both natural and clamped spline suffer equal error in the $[-2,2]$ range, except at specific points where they match $f(x)$ exactly. Chebyshev reduces all error near the sides but not near center.

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4. Approx. a function like $f(x) = \sin(10x)$ on $[0, 2\pi]$.

To make these splines periodic, I propose that the end point conditions at the conclusion of each period match those of the next.



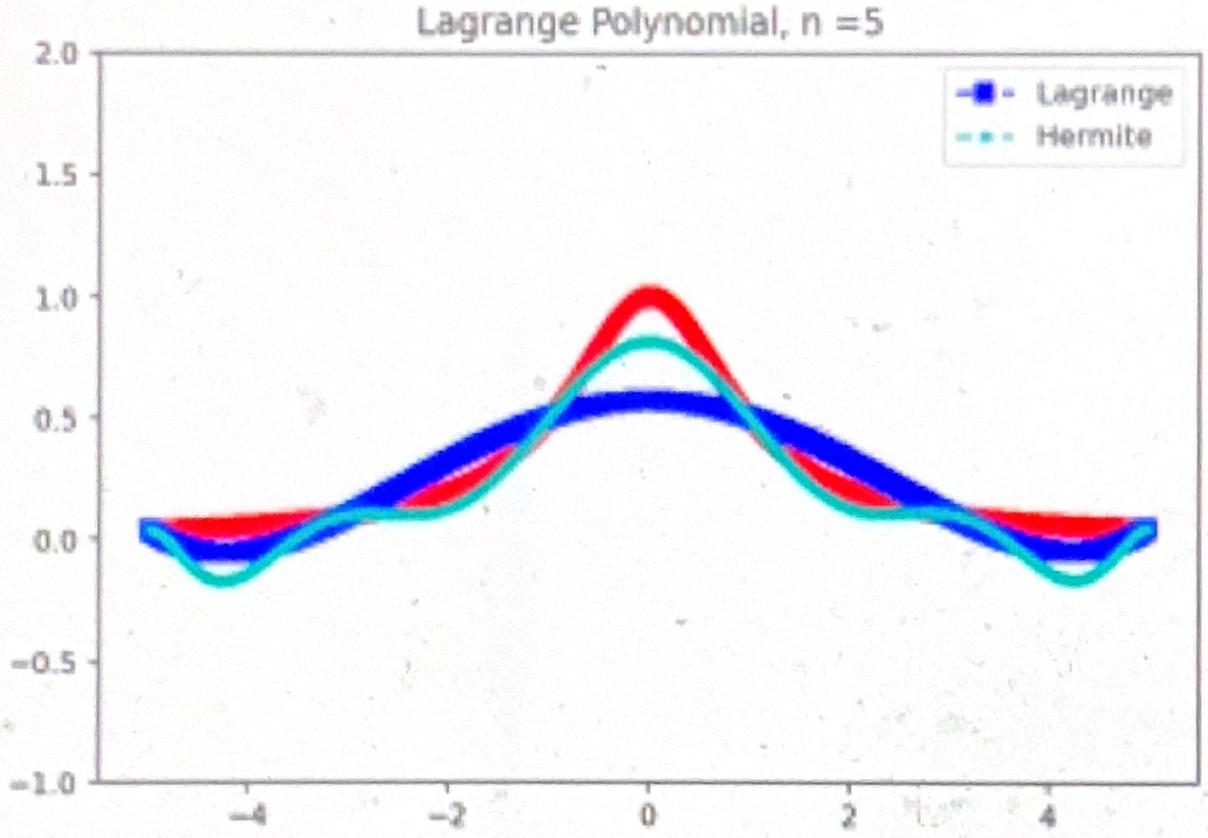
P = period of $S(x_n) = S(x_n + lP)$

This would look like:

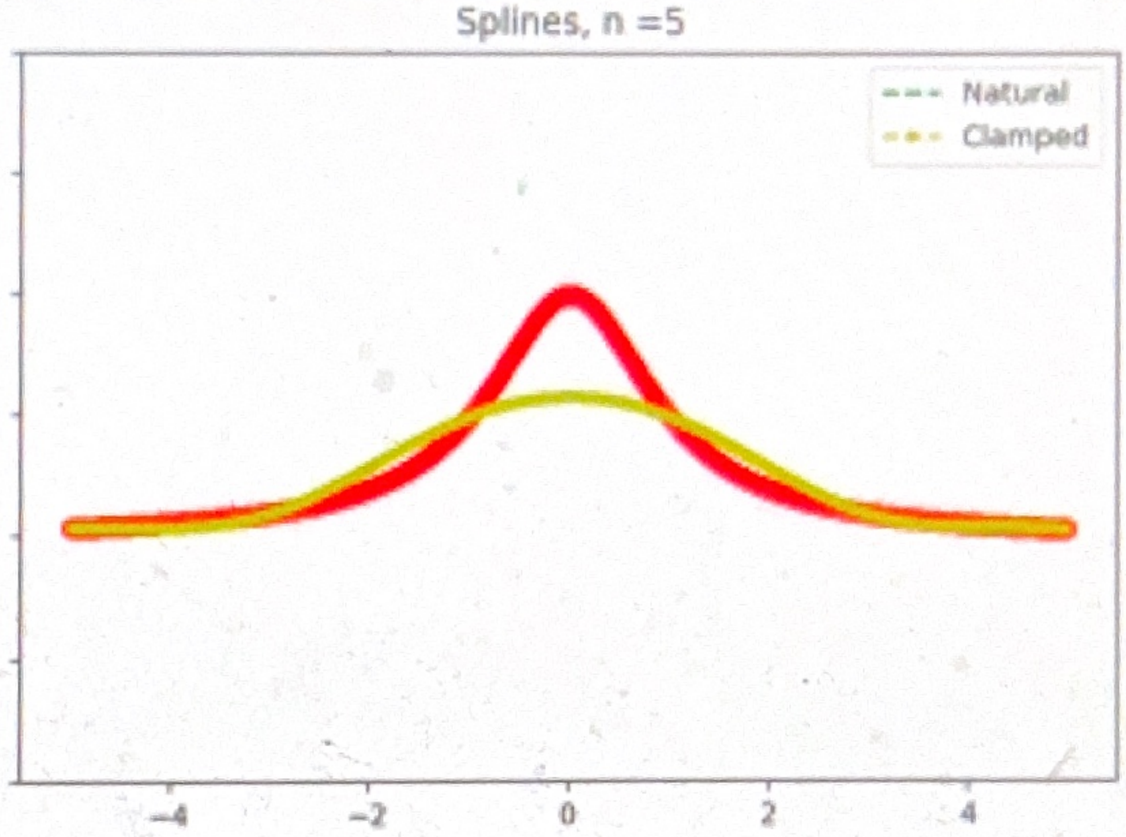
$$S_{n-1}(x_n) = S_{n+1}(x_n + lP)$$
$$S'_{n-1}(x_n) = S'_{n+1}(x_n + lP)$$
$$S''_{n-1}(x_n) = S''_{n+1}(x_n + lP) = M_{0,1} \neq 0$$

As a project I will code this.

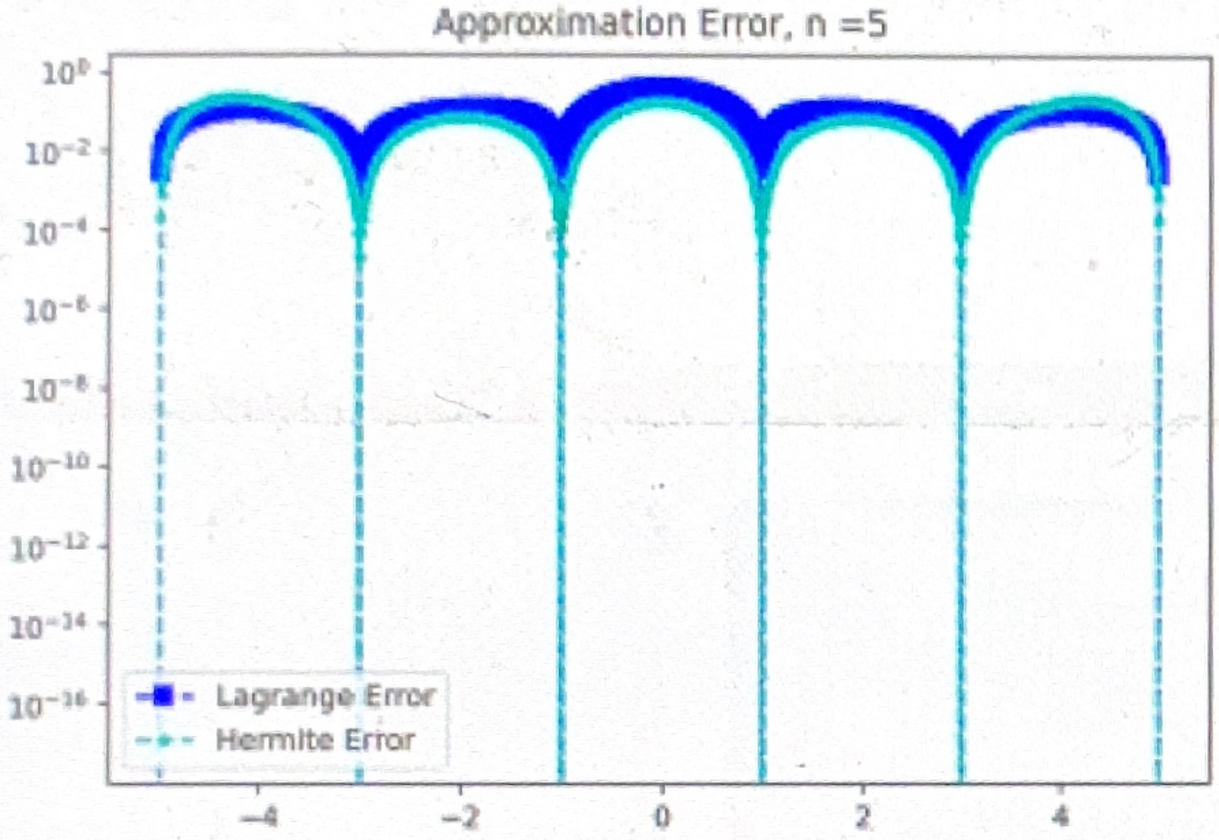
Lagrange Polynomial, n =5



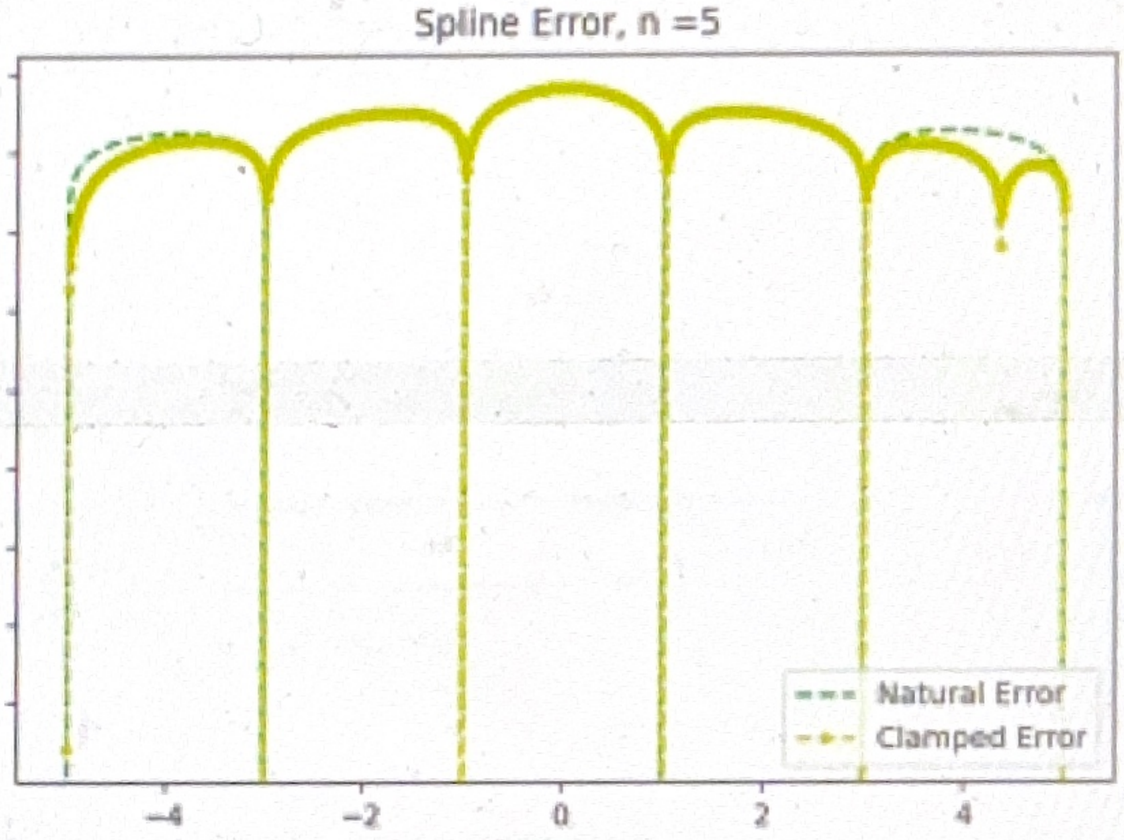
Splines, n =5



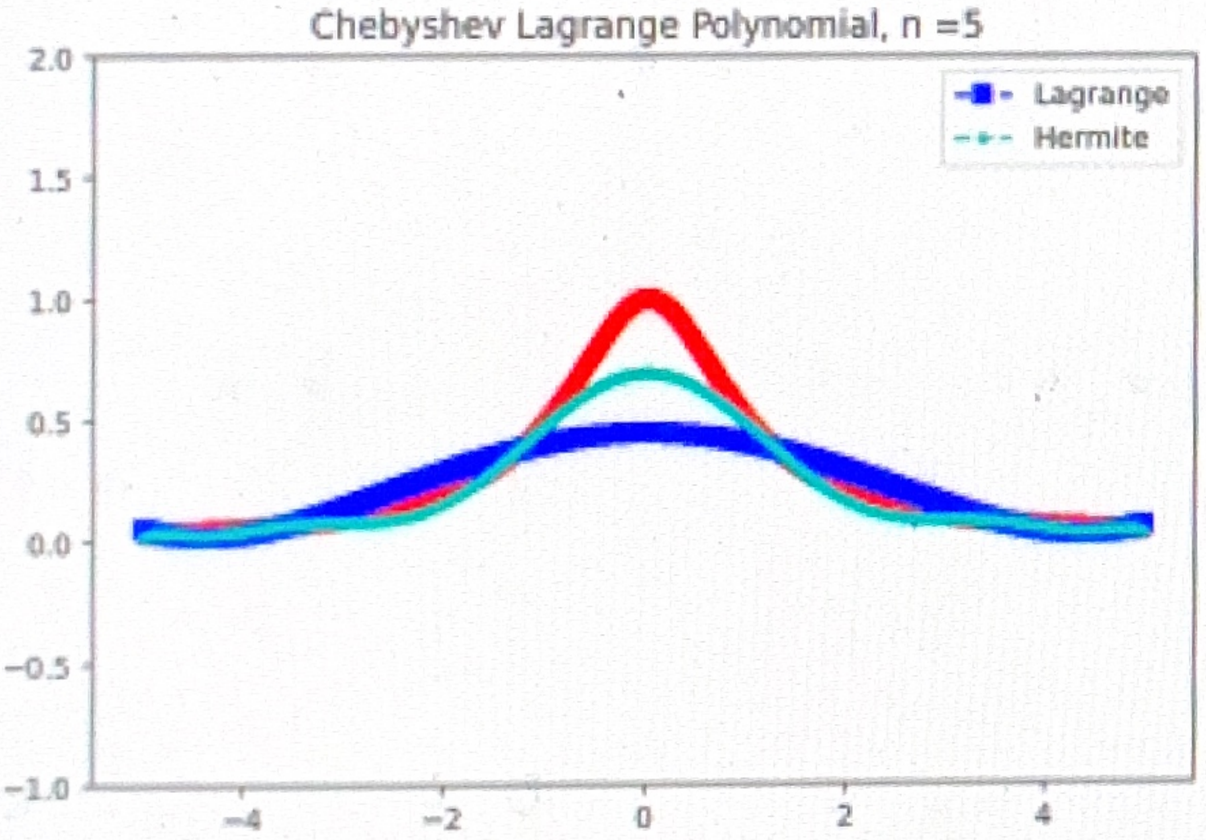
Approximation Error, n =5



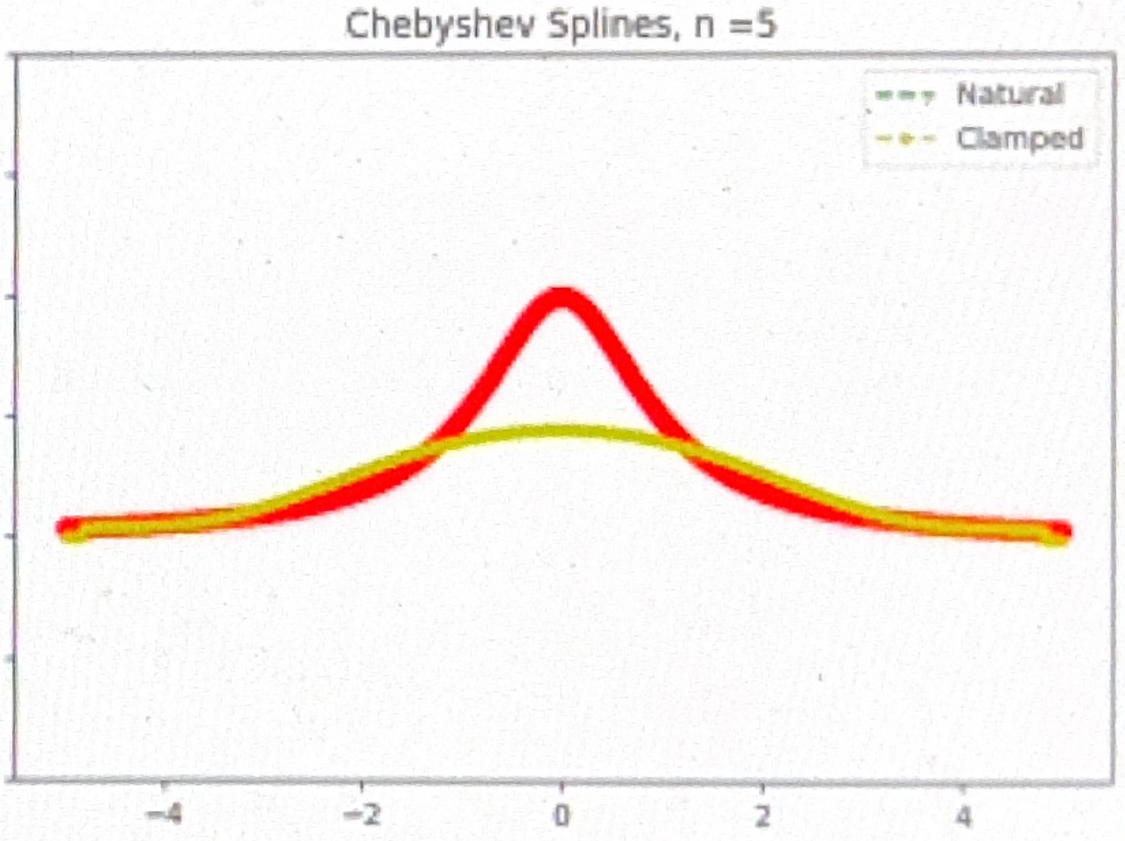
Spline Error, n =5



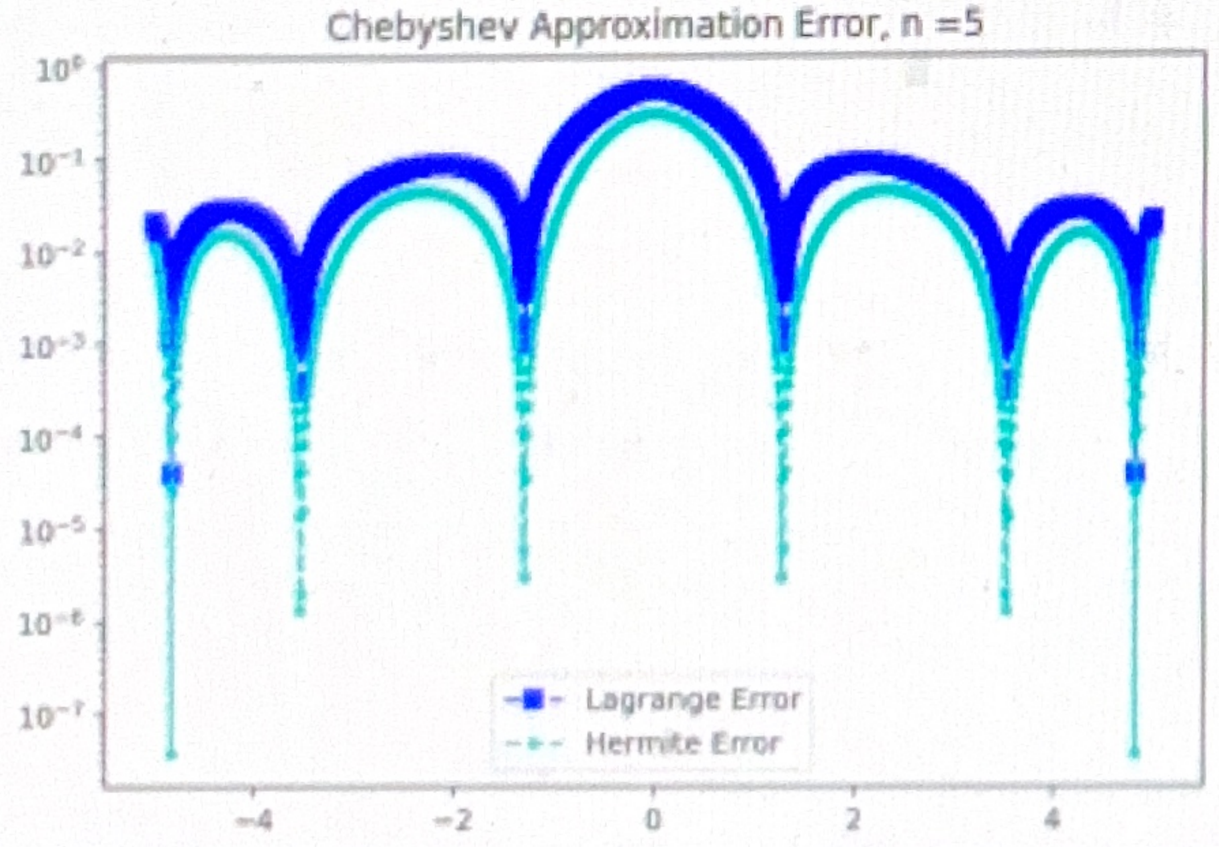
Chebyshev Lagrange Polynomial, n =5



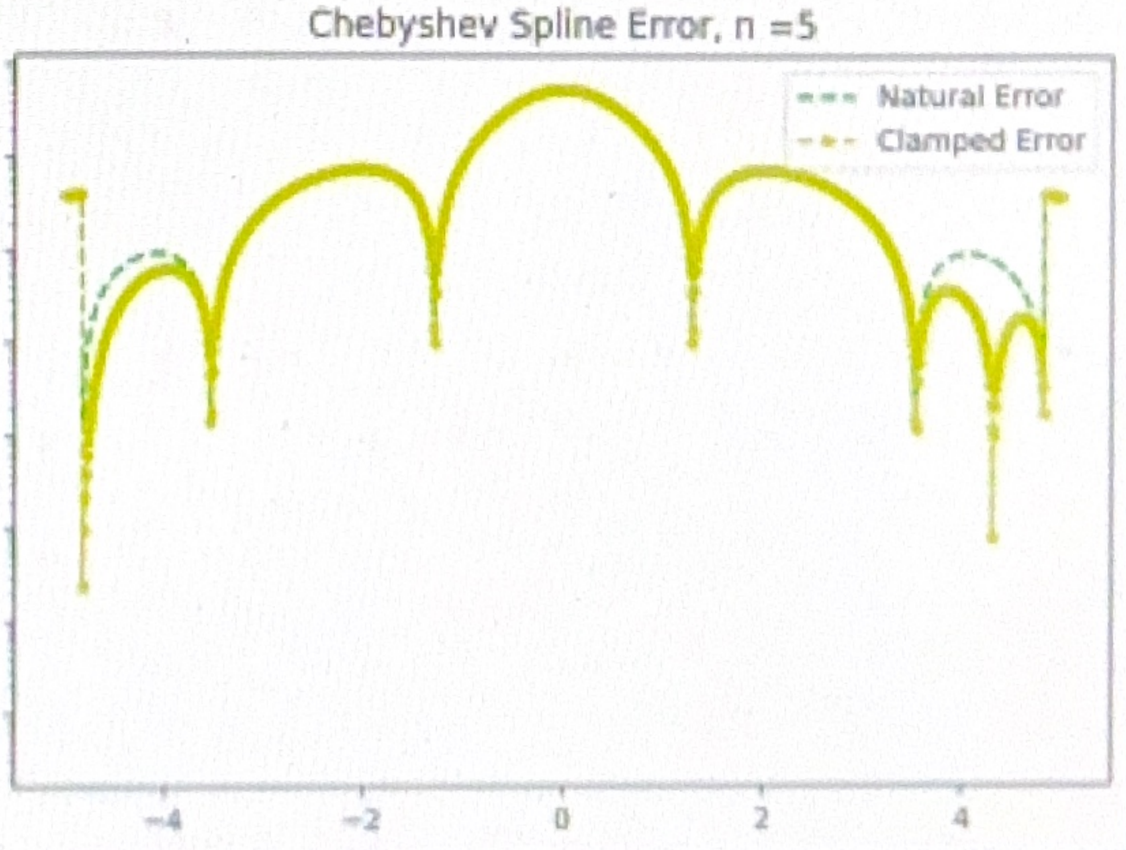
Chebyshev Splines, n =5



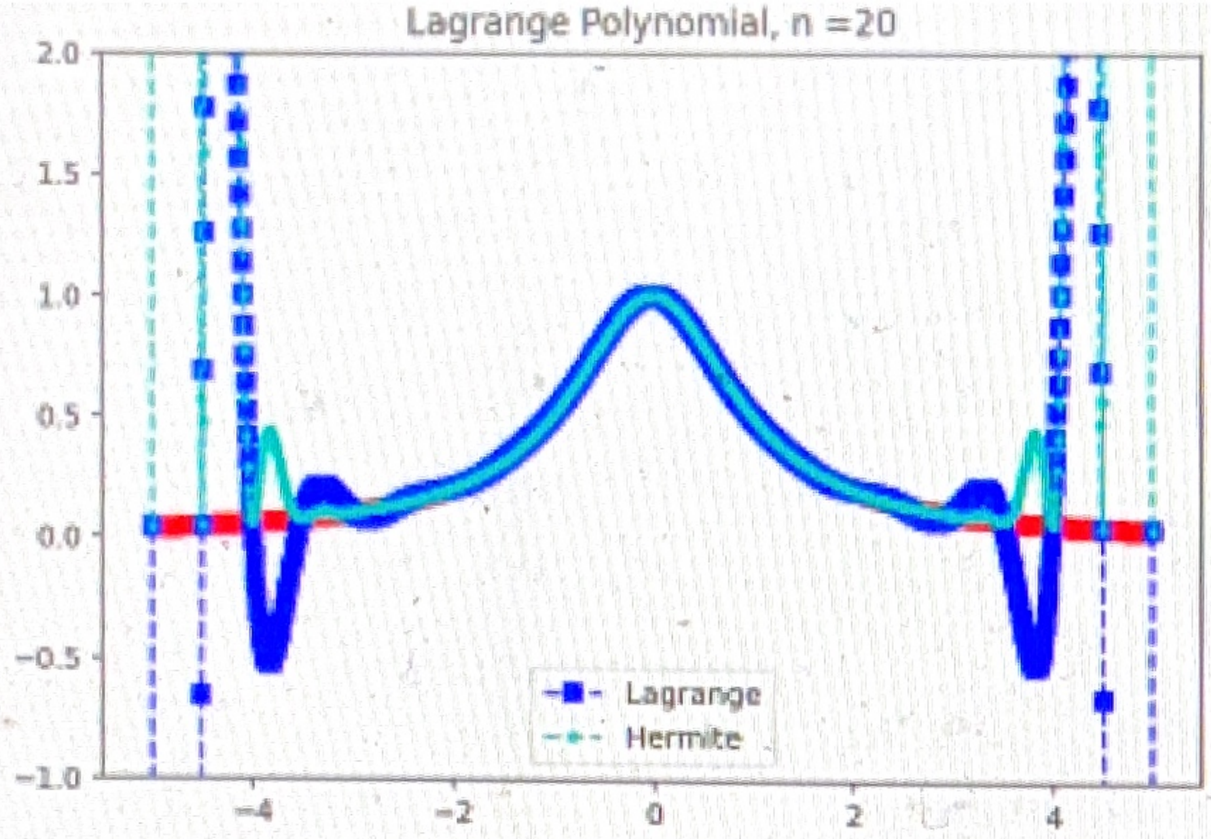
Chebyshev Approximation Error, n =5



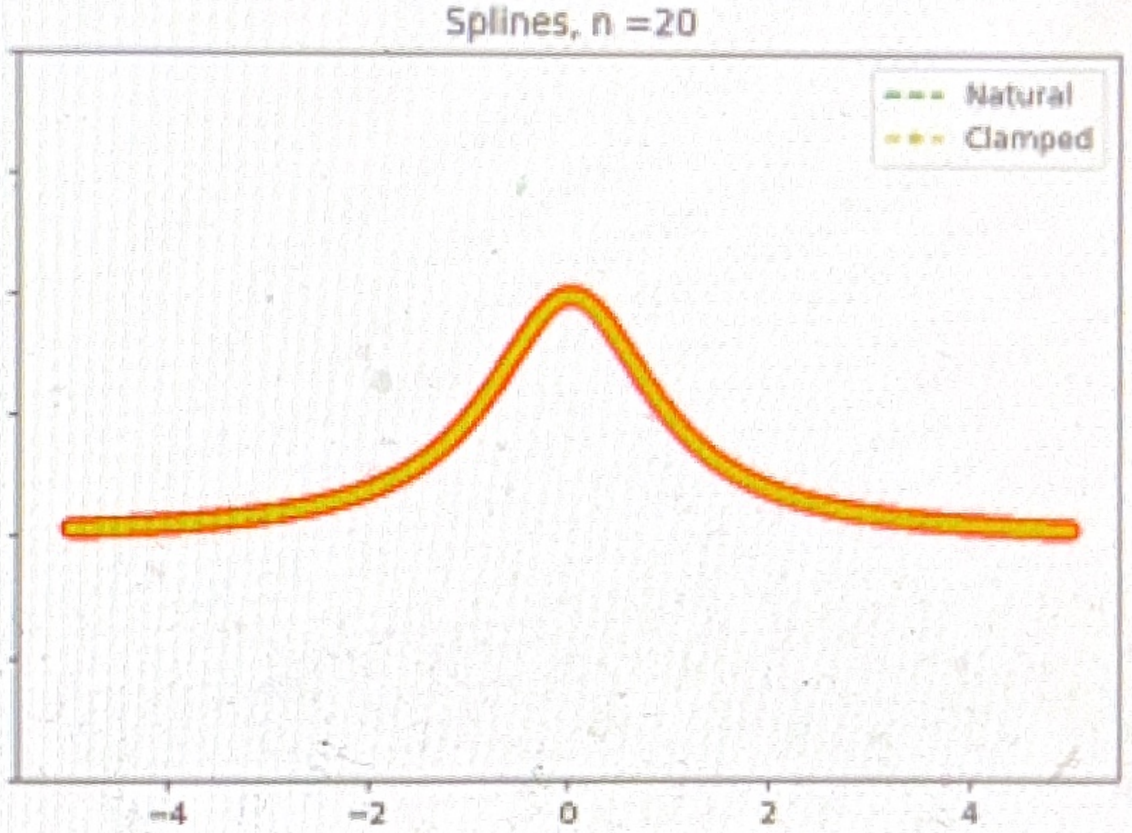
Chebyshev Spline Error, n =5



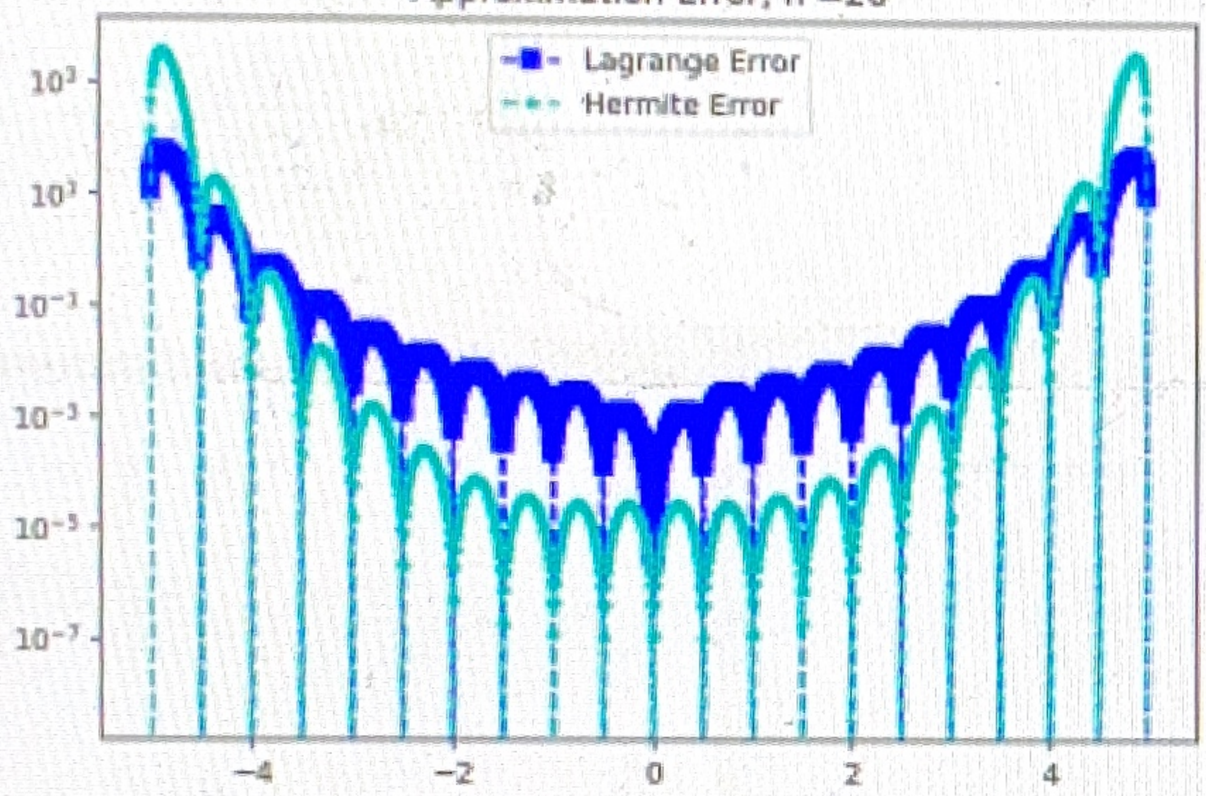
Lagrange Polynomial, n =20



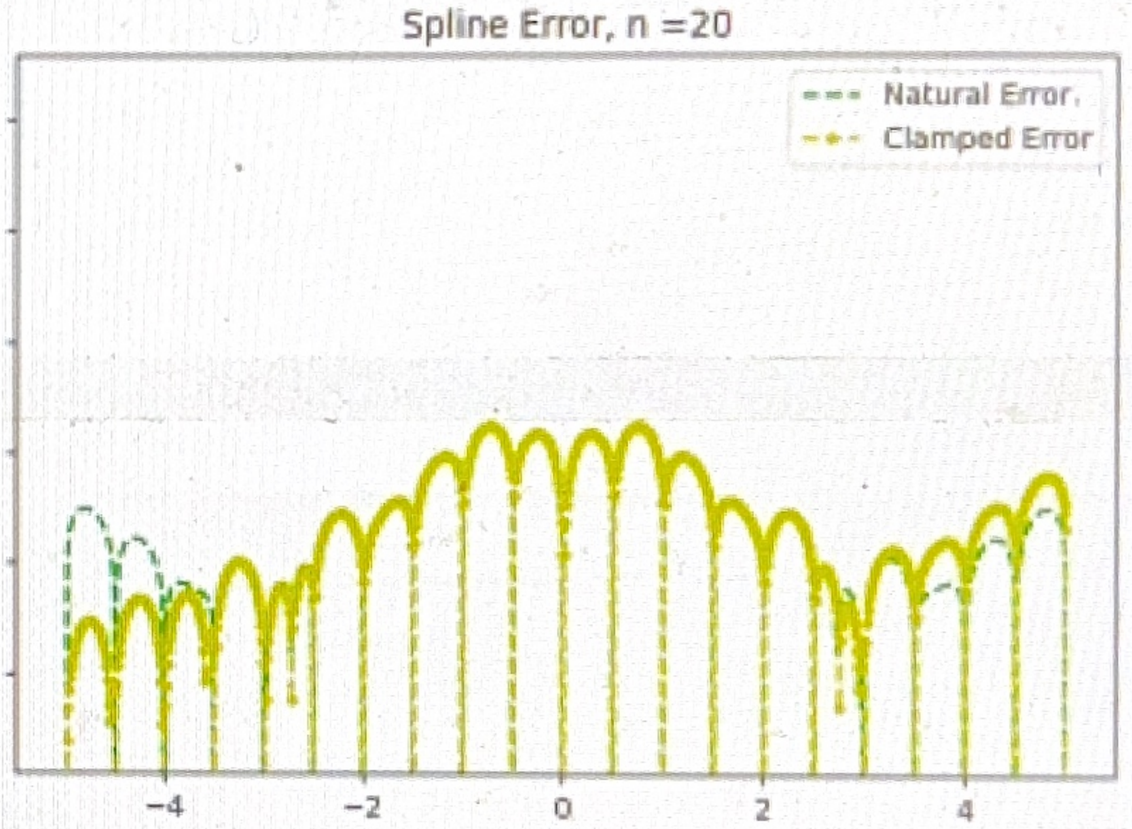
Splines, n =20



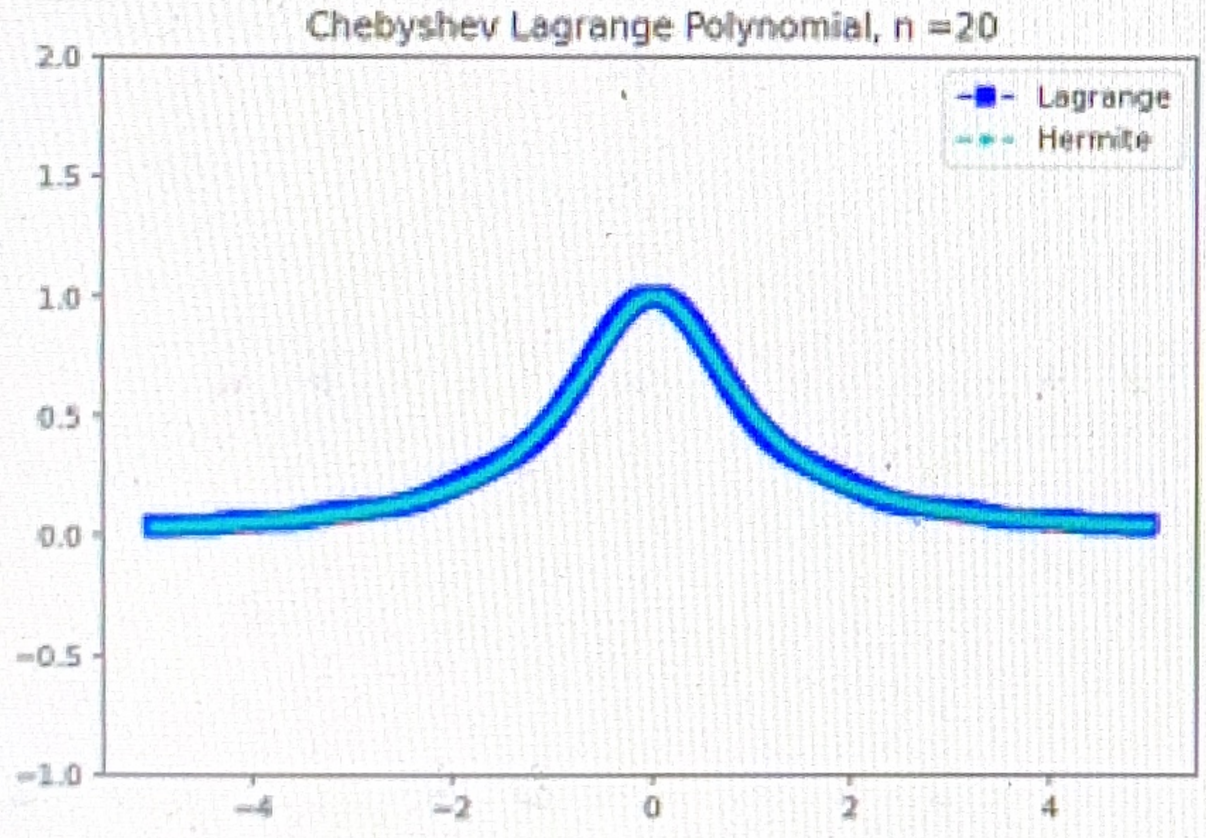
Approximation Error, n =20



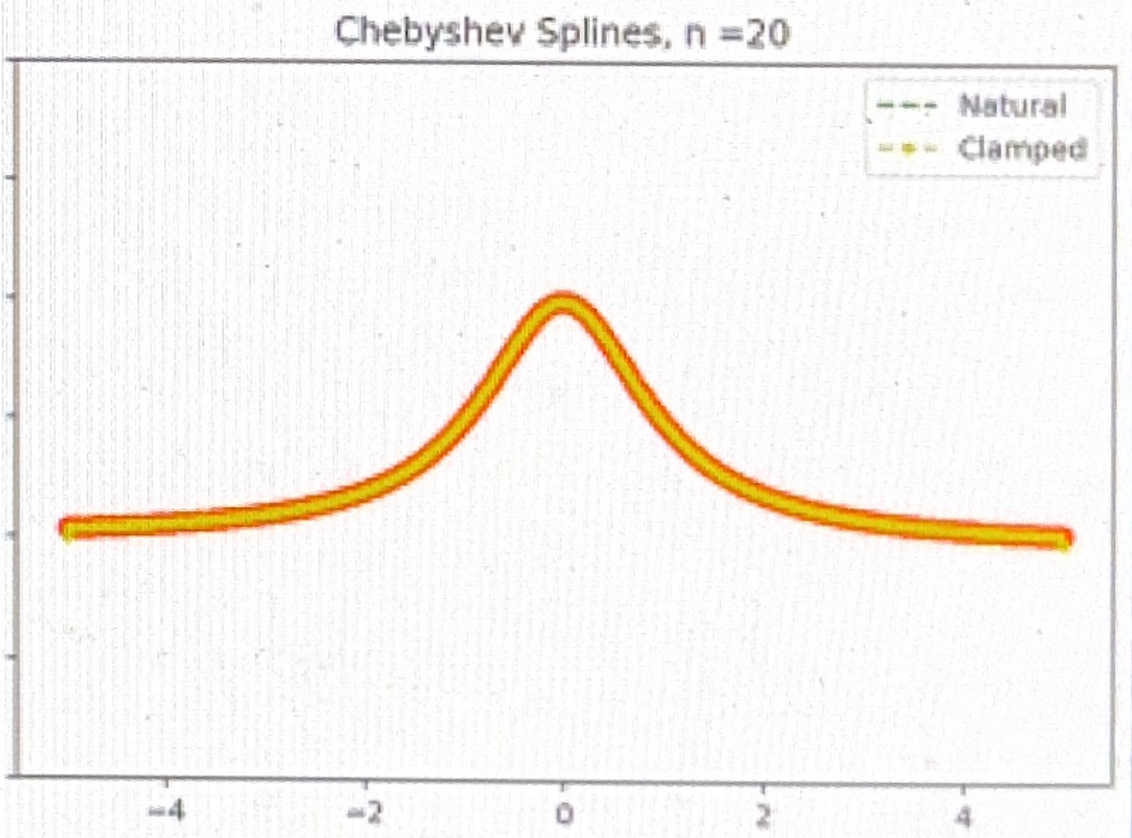
Spline Error, n =20



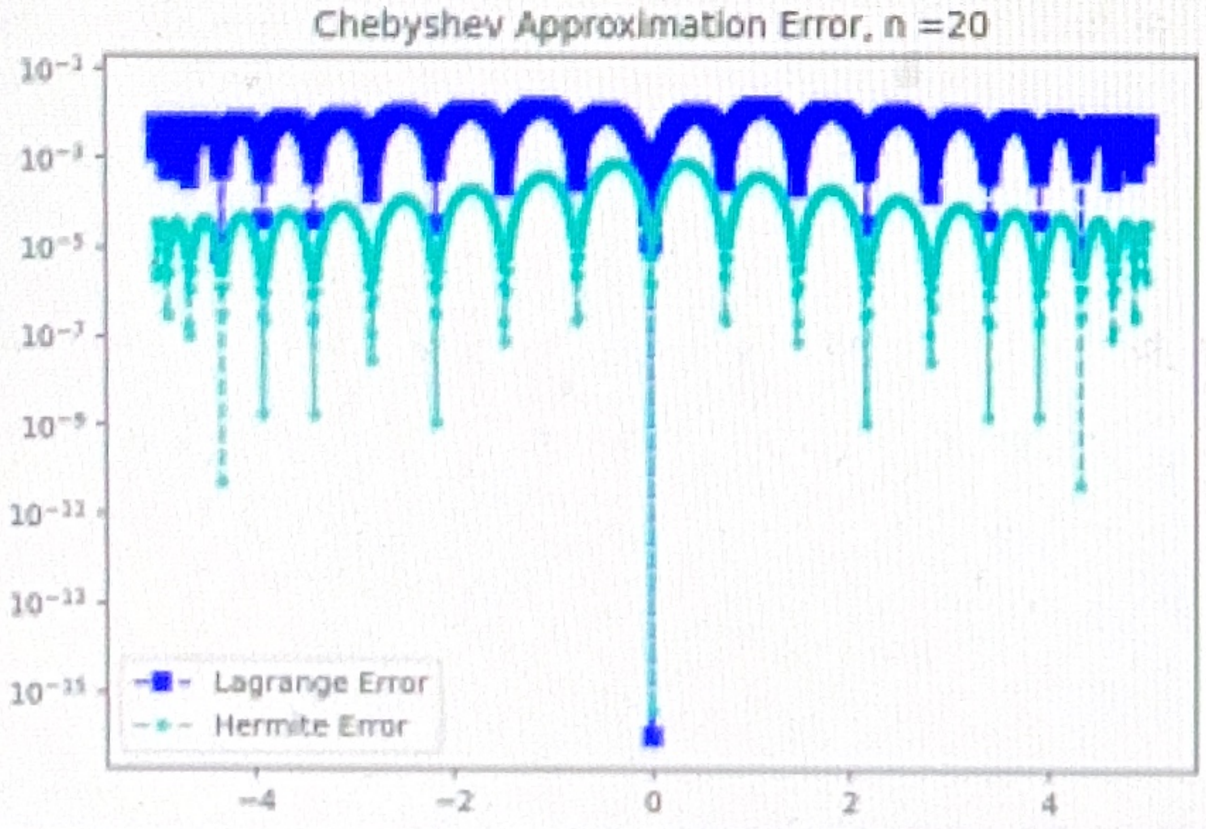
Chebyshev Lagrange Polynomial, n =20



Chebyshev Splines, n =20



Chebyshev Approximation Error, n =20



Chebyshev Spline Error, n =20

